Evolution of human intelligence

N. J. Mackintosh

The Dragons of Eden: Speculations on the Evolution of Human Intelligence. By Carl Sagan. Pp. 263. (Hodder and Stoughton; London, 1978.) £5.95.

CARL SAGAN is a professor of astronomy at Cornell and author or editor of several books such as *The Cosmic Connection: An Extraterrestrial Perspective*, and *Communication with Extraterrestrial Intelligence*. In *The Dragons of Eden*, he turns his attention to our own planet, and has written a personal, speculative account (his words, not mine) of the evolution of human intelligence. The new book has had great success in the United States (Random House: New York, 1977, $8.95; Ballantine: New York, 1978, paperback $2.25), and arrives here amidst much publicity. The publishers' blurb promises us "a breathtaking overview" containing "quite literally mind-blowing revelations" (literally?) about "how human beings evolved, brains and genes together".

The style is breathless, if not breathtaking. "Somewhere in the steaming jungles of the Carboniferous Period there emerged an organism that for the first time in the history of the world had more information in its brain than in its genes". Or "The R-complex [an entity of which more anon] is functioning in the dreams of humans; the dragons can be heard, hissing and rasping, and the dinosaurs thunder still". Sagan is much given to wonder as well as to purple passages. He wonders whether "the ritual aspects of many psychotic illnesses" and "the frequent ritualistic behaviour in young children" (what young children has he been watching?) represent a failure of neocortical control. And noting that australopithecines were only three-and-a-half feet tall, "I wonder," he tells us, "whether our myths about gnomes, trolls, giants and dwarfs could possibly be a genetic or cultural memory of those times".

This, then, is a popular book, written by a non-specialist straying outside his academic fold. If it has some of the virtues of its kind, being reasonably free of jargon, easy going, occasionally interesting, and often nice to look at, this is amply compensated for by its vices. It is inaccurate, full of fanciful and unilluminating analogies, infuriatingly unsystematic, and skims hither and yon over the surface of the subject, unerringly concentrating on the superficial and misleading at the expense of the important or significant. That it should have had such success is a sad commentary on the power of publicity.

The book is well produced and easy on the eye. There are numerous illustrations of tangential relevance to the text. Sagan's favourite artist seems to be Escher, but others who have painted or sculpted serpents and dragons are not neglected. The text itself contains some quaint, if not particularly useful, snippets of information. I did not know, for example, that Lord Byron's brain was twice the size of Anatole France's, nor that Pliny had supposed the ostrich to be the result of a cross between a giraffe and a gnat (can he really have thought that?). Nor had I come across the notion that our mistrust of left-handedness (sinister, gauche), and preference for using the right hand in greeting might have arisen from our ancestors' habit of using the left hand "for personal hygiene after defecation".

But these are not great virtues. The inaccuracies alone suffice to negate them, especially when they come as thick and fast as in this single sentence: "A fish has a notochord or spinal cord, which it shares with even humbler invertebrates". And all too often the snippets of information distract rather than inform. Does it help one understand what an average brain volume of 1375 ml means to be told that 1 ml is about the volume of an adult human navel? At times, indeed, the book degenerates into a series of random odds and ends, strung together without rhyme or reason. One chapter, for example, entitled "Tales of Dim Eden", starts with some information about brain waves, shows us the electroencephalogram patterns characteristic of waking, sleeping and dreaming, and provides some speculations about the functions of sleep. We then move on to a discussion of the brain size of dinosaurs, some speculations on the causes of their extinction, and the suggestion that the evolutionary conflict between mammals and reptiles is mirrored in the story of the serpent in Eden. We read about the Komodo dragon of Indonesia, "a predator exhibiting a chilly fixity of purpose", and are told that stories of dragons are common in all mythology and folk tales. We then return to dreams and their functions, are asked to believe that "major evolution beyond the reptiles has been accompanied by and perhaps requires dreams", and are finally informed that the true function of dreams is to allow free play to the ancient, reptilian parts of our brain (the R complex). Notwithstanding which, dreams about falling are said to reflect our arboreal origins as primates, and those about snakes our mammalian fear of reptiles.

If he perseveres to the end, therefore, the innocent reader will have supped on a random array of information, alternately entertaining and bizarre, will have felt his flesh creep with stories of dragons, and will conclude with some mildly uplifting exhortations to trust to reason and science rather than the occult or irrational—a nice touch this in a book so profoundly unscientific. Will he have learned anything about the evolution and nature of human intelligence? Not much. It is a pity. It is an important and fascinating topic. We all, no doubt, want to know more about ourselves. What manner of animal are we and where do we come from? What makes us different from other animals and what do we mean when we say we are more intelligent than them? But we must temper our fascination with some respect for hard evidence, and therein lies the problem, for in the nature of the case these are questions about which it is easier to speculate than provide such evidence. From palaeontology we know that the evolution of the human brain followed rather than preceded the development of walking on two legs, and that early hominids made tools and ate meat. We can thus speculate that human intelligence developed in response to tool making and cooperative hunting, but the inference is by no means as plausible as is popularly supposed. One need only, for example, look at chimpanzees to see that walking on one's hind legs is hardly a

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necessary precursor to using one's hands to manipulate or fashion implements. And what is the relationship between the brain and intelligence anyway? That there is *some* relationship between the size of one and the degree of the other seems probable. But it is surely not as close as is sometimes assumed, and one suspects that the use of brain size as an index of an animal's intelligence is more a consequence of the lack of any better measure rather than a reflection of its intrinsic worth. Sagan is by turns sensible and rash on this topic. He starts cautiously, but in a few pages one reads, for example, that "the evidence is quite compelling that mammals are indeed systematically more intelligent than reptiles". The only evidence mentioned (unsurprisingly, as there is no other available) is that the ratio of brain weight to body weight in mammals is significantly higher than in reptiles, but this by itself proves nothing.

In common with others who should know better, Sagan is prone to naive neurologising. The central thesis of his book, indeed, is a classic example of this tendency. Because parts of the human brain are supposedly homologous to parts of the brain of reptiles, Sagan jumps to the unwarranted conclusion that they continue to perform the same, reptilian functions. And a sinister set of functions they are: aggression, territoriality, hierarchy, and ritual. Of these, his favourite is ritual—perhaps because of the alliterative delights of talking of the ritualistic, reptilian functions of the R complex. At any rate, the desire to dress up, the belief in sacred rites, and the compulsion to obey mysterious laws, are all a consequence of our reptilian ancestry. Lip service is paid to our ability to overcome these primitive impulses (through neocortical control), but the dark hint is there. Not that it lacks a lighter side: I particularly liked the "reptilian ritualization of the educational process".

The brain is a physical entity, and those who need to see a piece of machinery before they can accept that it performs certain functions will always be tempted by facile explanations of psychological process in terms of neurological structure. As neither psychological process nor behaviour leave fossil evidence, the temptation is all the more irresistible when we ask questions about the evolution of behaviour. Brains themselves, of course, do not fossilise, but skulls do; from fossil skulls we can reconstruct endocasts, and from these we can make more or less informed guesses about the size of the brain of some extinct ancestor of man. It is important to realise how extraordinarily little this will have told us about his behaviour. Perhaps we would do better to study the behaviour of those living species most closely related to man. Sagan does not neglect this alternative, and takes us on a quick tour of some of the more widely popularised features of monkey and chimpanzee life. We learn of Japanese macaques learning to unwrap caramels and sifting wheat from sand, and transmitting these new habits to other members of the troop; of chimpanzees fishing for termites in East Africa, and learning sign language in Nevada. Every example is treated with bated breath. Together, they suggest to Sagan that within a few generations we may see a community of chimpanzees from which "might emerge the memoirs of the natural history and mental life of a chimpanzee, published in English or Japanese (with perhaps an 'as told to' after the by-line)". Whether or not Sagan wishes us to take this possibility seriously, it remains entirely typical that he nowhere considers just what intellectual or cognitive capacities are implied by the feats over which he enthuses. Learning to fish for termites, it seems, is very much a matter of trial and error, there being no way to predict from rational considerations what sort of twig is best adapted to the task, nor how to twiddle it around, nor even where to put it, and young chimpanzees learn these skills largely by careful observation of their elders. Neither trial and error learning nor social imitation are particularly impressive examples of higher intellectual processes, and neither, of course, is confined to primates. Although Sagan tells us how marvellous it is that chimpanzees can be taught to communicate by a sort of gestural language, he does not discuss what is involved in this behaviour, nor how it differs from human language; nor does he even mention the work of David Premack, the one investigator of chimpanzee 'language' who has attempted to understand some of the cognitive processes and skills underlying it.

Characteristically, it is the fanciful and arresting implication that catches Sagan's attention. Thus, "if chimpanzees have consciousness, if they are capable of abstractions, do they not have what until now has been described as 'human rights'? How smart does a chimpanzee have to be before killing him constitutes murder?". That is a question worth considering, but it seems only fair to conclude with a more typical Saganism. If "there are non-human primates so close to the edge of language, so willing to learn, so entirely competent in its use and inventive in its application . . . why are they all on the edge? Why are there no non-human primates with an *existing* complex gestural language?" The answer? It is because "humans have systematically exterminated those other primates who displayed signs of intelligence". □

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